

XRPL Validator — Daily Engineering Report • 2026-09-23 (in progress)

Generated 2026-09-23 16:06 EDT. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the live native-engine shadow, the reference rippled node, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **8 commits** (xrpl-validator 8)
- **14 engine findings fixed** (F383–F396)
- 3 test / vector commits
- Live shadow: 451/451 ledgers fully matched, 42,096 transactions, 0 engine receipts in the current soak
- Amendment **PermissionDelegationV1_1** lost majority at ledger 107,181,569 (2026-09-23 12:47 UTC)
- Amendment **BatchV1_1** held majority (since 2026-09-15 14:06 UTC); activates ~2026-09-29 14:06 UTC

Live validator and native engine

Snapshot at generation time.

Measure	Value
Ledgers compared (current run)	451
Ledgers fully matched	451
Transactions applied natively	42,096
Result mismatches	0
Byte mismatches	0
Missing / extra keys	0 / 0
Batch inner-result mismatches	0
Engine receipts (current soak)	0
Live engine build running since	2026-09-23 15:30 EDT
Reference rippled	3.4.0, full, validated #107,188,382

Engine findings fixed

- **fix(ledger): F383-F388 — AMM deposit/withdraw sub-modes ported from rippled 3.4.0 (campaign 16)**
(4789c7d, xrpl-validator) Testnet campaign 16 (AMM deposit/withdraw sub-modes against rippled 3.4.0) found six rules the engine lacked or got wrong: 16 of 138 bundles failed under BOTH engines (the transactors are shared). Each fix is a structural port — rippled's function body line for line, in its operation order and Number rounding modes, with its result codes in its order — on the signed Number port (tx::number) where the unsigned n_* ...

- **fix(ledger): F389-F394 — Batch inners: common preclaim, delegate permission, AccountTxnID order, AccountDelete fee, per-inner threading (campaign 23)** (74f5bbd , xrpl-validator) Campaign 23 (devnet, 2026-09-23) crossed Batch (BatchV1_1 activates on mainnet ~09-29) with delegation, MPT, credentials, tickets, AccountDelete and LedgerStateFix inners. Six rules the engine had wrong: F389 (6-2, 6-5, 6-6, 6-7, 6-9) — a Batch inner runs rippled's whole per-transaction pipeline, common preclaim included (checkSeqProxy, checkPriorTxAndLastLedger — applySteps.cpp:179-183): an inner with a future/past ...
- **fix(ledger): F395-F396 — Batch touched sets read decoded pre-images; inner ids use the shared canonicaliser** (f7ad693 , xrpl-validator) A fresh-context review of F389-F394 against rippled 3.4.0 found two real problems (no critical one in the live-shadow path): F395 — tx/batch.rs built each inner's touched set with the pre-batch value from `state_map.lookup`, the RAW leaf. `state_replay` stores its leaves binary behind a `leaf_decoder`, so binary was compared with the sandbox's JSON: every unchanged write-back of a base object looked like a change (a ...

Tests and campaigns

- **test(node): campaign 23 — 60 byte-exact Batch × feature vectors; fetcher read-sets, build_batch.py** (e142bc1 , xrpl-validator)
- **test(node): probe_bundle judges a Batch whole — per-inner threading and inner verdicts** (a05e2dc , xrpl-validator)
- **test(node): campaign 24 — Batch × NFT, offer, check, channel, escrow, oracle, DID, AMM, trust line, AccountSet (27 vectors)** (ba89585 , xrpl-validator)

Docs and operations

- **docs: README — live-shadow status and progress history** (7857d61 , xrpl-validator)
- **docs: README — second clean 24-hour run; link the daily reports** (863ec8f , xrpl-validator)

Notes

Hand-written notes on the day's XRPL work: research and operations.

- **PermissionDelegationV1_1 lost majority on mainnet.** The network applied `EnableAmendment` with `tfLostMajority` at ledger 107,181,569 (2026-09-23 12:47 UTC). It had held a majority since 2026-09-21 and would have activated around 2026-10-05; the two-week clock restarts only if it regains majority. **BatchV1_1 still holds its majority** (since 2026-09-15 14:06 UTC) and activates around 2026-09-29 14:06 UTC.
- **The second clean 24-hour run.** The live soak started 2026-09-22 11:38 EDT passed its 24-hour mark with zero engine receipts — every compared ledger matched rippled byte-for-byte (the first such run was 2026-09-20/21).
- **Batch conformance ahead of activation.** Campaign 23 (devnet) crossed Batch with delegation, MPT, credentials, tickets, AccountDelete and LedgerStateFix inners: six engine rules fixed (F389–F394). The subtlest: rippled threads an owner's AccountRoot (`threadOwners`) only for nodes an inner itself inserted or erased, in that inner's own ApplyStateTable — not for whatever the batch as a whole created. Campaign 24 (devnet) put NFT,

offer, check, payment-channel, escrow, oracle, DID, AMM, trust-line and AccountSet inners inside Batches, each shaped so a later inner uses what an earlier one created: 27/27 byte-exact on the first run.

- **An independent review against the rippled 3.4.0 sources** of the Batch fixes found two more issues, fixed test-first (F395, F396): a replay-only pre-image read of raw leaves, and inner transaction ids hashed without the codec's canonical spelling of decimal MPT amounts and short-hex `AssetPrice`.
- **Desk checks for the two Batch-only bug classes found so far:** no other transactor copies raw transaction `Flags` into a ledger object (where `tfInnerBatchTxn` would leak), and none gates the fee level in preclaim (a Batch inner carries `Fee: "0"`).
- **Deployment.** The engine build f7ad693 (findings through F396) replaced the live shadow build at 15:31 EDT after its gate (1,589 suite tests and 706 flow-engine tests passing) and two mainnet replay windows (60 and 98 ledgers, zero mismatches). The post-deployment health check matched 402 ledgers with no mismatches, and a new 24-hour soak began at 15:57 EDT (seed ledger 107,187,825).
- **Receipt canary — first live run.** One drop was planted in the shadow's copy of a mainnet AccountRoot at ledger 107,188,304; the byte-for-byte compare flagged it and the receipt watcher logged it 255 seconds later. Canary lines are never counted as receipts: the check exists to prove that a zero-receipt soak means the comparison was running and sensitive. It now runs daily.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the latest validated ledger, and the day's EnableAmendment pseudo-transactions.

Voting events:

- `PermissionDelegationV1_1` lost majority — ledger 107,181,569, 2026-09-23 12:47 UTC

Amendment	Majority since	Activates (if held)
BatchV1_1	2026-09-15 14:06 UTC	2026-09-29 14:06 UTC

Amendments enabled on mainnet: 94.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Hub-Mediated Multilateral Check Netting** (settlement) Add transaction `ttCHECK_NET` carrying `sfCheckIDs` (`STVector256`, bounded ≤ 8) referencing existing Check objects where the submitting account is either `sf`
- **On-Ledger Settlement Finality Receipts** (settlement) Add an optional `tfFinalityReceipt` flag to `EscrowFinish` and `CheckCash`; when set and the discharge succeeds, `doApply` additionally creates a compact `ltSE`
- **Destination-Enforced Travel-Rule Envelope** (compliance) Add `lsfRequireTravelRule` (AccountSet-settable flag) and `sfTravelRuleThreshold` to AccountRoot, plus a new bounded VL SField `sfTravelRuleEnvelope` (max 4

Appendix — commits

Time	Repo	Commit	Subject
01:54	xrpl-validator	4789c7d	fix(ledger): F383-F388 — AMM deposit/withdraw sub-modes ported from rippled 3.4.0 (campaign 16)
02:57	xrpl-validator	e142bc1	test(node): campaign 23 — 60 byte-exact Batch × feature vectors; fetcher read-sets, build_batch.py
02:57	xrpl-validator	74f5bbd	fix(ledger): F389-F394 — Batch inners: common preclaim, delegate permission, AccountTxnID order, AccountDelete fee, per-inner threading (campaign 23)
03:15	xrpl-validator	a05e2dc	test(node): probe_bundle judges a Batch whole — per-inner threading and inner verdicts
03:47	xrpl-validator	ba89585	test(node): campaign 24 — Batch × NFT, offer, check, channel, escrow, oracle, DID, AMM, trust line, AccountSet (27 vectors)
04:21	xrpl-validator	f7ad693	fix(ledger): F395-F396 — Batch touched sets read decoded pre-images; inner ids use the shared canonicaliser
10:42	xrpl-validator	7857d61	docs: README — live-shadow status and progress history
12:25	xrpl-validator	863ec8f	docs: README — second clean 24-hour run; link the daily reports